
Core Concepts in Quantum Field Theory

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This note is a summary of formal core concepts in quantum field theory. Including Symmetry, Renormalization, Anomalies, Solitons and Instantons. The note is mainly based on Peskin and Schroeder's "An Introduction to Quantum Field Theory"[1], Tong's lecture notes on gauge theory, and some other references.

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CHAPTER 1

Renormalization

1 Symmetry and Ward-Takahashi Identities

Anomalies

1 Chiral Anomalies

This section I will follow David Tong's lecture on gauge theory to talk about chiral anomalies in fermions when coupled to gauge fields.

Symmetry Breaking

Bibliography

- [1] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory*. Reading, Mass: Addison-Wesley Pub. Co, 1995.